



Recent Projects

2026

Treasury Caelum EKS

U.S. Department of the Treasury, Washington, DC

Provided technical leadership for a cross-team effort to build out a shared application deployment platform on Amazon EKS. The team built out private-cluster access, managed compute, GitOps reconciliation, networking, workload identity, rebuild, and operational workflows.

2025

The Charge to Timothy & Titus

Personal Project

I set the NASB 1977 text of I Timothy, II Timothy, and Titus to music in lead sheet format. I leveraged AI tools to create software which generated score videos of the songs, and created a printed booklet of the songs with QR codes that link to the videos on YouTube.

2025

go-mem

Personal Project

In order to teach myself some things about the Go programming language as well as AI tools, I built `go-mem`, a console mode game to assist in memorizing texts. The project may be viewed and downloaded at <https://github.com/ArkieCoder/go-mem>.

2025

AWS/Kubernetes Cluster Compliance

SPS Commerce, Rogers, AR

Implemented a comprehensive labeling scheme in our Kubernetes cluster to comply with SPS Commerce's post-acquisition requirements, along with a tagging regime required by SPS with regard to our AWS infrastructure. Additionally, architected the solution to switch Kubernetes secrets from HashiCorp Vault secrets where Vault ran within our clusters, to using AWS Secrets Manager where the secrets existed in a different AWS account. This required careful cross-account IAM design and coordination across teams.

2025

Move to One URL

SPS Commerce, Rogers, AR

SupplyPike, a subsidiary of SPS Commerce, originally had multiple app URLs. Using a combination of the Traefik reverse proxy server for Kubernetes and configuration of Amazon Application Load Balancers, I successfully switched SupplyPike from using multiple application domains to just one. This required careful coordination over weeks with different teams, and planning how things were rolled out.

2023-2024

Data Warehouse Ingestion Pipeline

AFG.tech, Dallas, TX

Architected a data ingestion pipeline using AWS Lambda and DynamoDB. This pipeline connected to the APIs of various data providers, ingested the raw data, converted it to JSON, and mapped the JSON fields coming from the data provider to the equivalent field in our DWH schema. Then, the data would be stored in an S3 bucket for ingestion into a Redshift cluster. The purpose of this project was to provide an incoming team of data engineers with a starting point for their efforts to make a truly resilient and configurable data pipeline that met business needs.

2022-2023

OpenSearch Domain Architecture

Simmons Foods, Siloam Springs, AR

Architected a SIEM stack in AWS using Fluentd for log ingestion and the AWS OpenSearch service for search and visualization. The AWS parts were built using Terraform, and the OpenSearch configuration was handled in an idempotent way by leveraging the OpenSearch APIs via shell scripts. Fluentd was implemented using three EC2 instances running behind a Network Load Balancer.

2022-2023

BeyondTrust PRA Implementation

Simmons Foods, Siloam Springs, AR

Worked on the rollout of BeyondTrust PRA (Privileged Remote Access) throughout the organization for the purpose of VPAM (Vendor Privilege Access Management). Leveraged published and unpublished APIs to create an Ansible plugin for configuring BeyondTrust, reducing the administrative burden to editing a series of YAML files.

2020-2021

Built Terraform Deployment Framework

Simmons Foods, Siloam Springs, AR

Created the `configure_deploy.py` script to configure the deployment of infrastructure defined via Terraform to easily deploy the code from any branch of any GitHub repository to any AWS account using any set of environment variables.

2020-2021

Defined Full Stack App Architecture as Code

Simmons Foods, Siloam Springs, AR

Using Terraform, created code to roll out: AWS shared objects (objects shared across full stack apps), Image Build Pipelines, EC2 instances, RDS instances, load balancers, S3 buckets, users, profiles, Code Pipelines, etc. This code was modularized so that all pieces of a full stack app could be actuated from one Terraform file.

2019-2020

Successfully Pushed for Adoption of Git, Then GitHub

Simmons Foods, Siloam Springs, AR

Spearheaded the use and adoption of Git source management tool internally, then led use and adoption of GitHub SaaS after it was determined that Git added value. I coordinated the migration of on-premise Git repositories to GitHub, and leveraged GitHub Actions to automatically deploy Microsoft SQL Server stored procedures, application configuration, and Oracle BI Publisher reports.

2019

Defined Full Stack App Strategy

Simmons Foods, Siloam Springs, AR

When I arrived at Simmons, the web application strategy revolved around Oracle Apex, a free, low-code platform tightly integrated with the Oracle Database. Initially, I worked to establish Git repositories (later GitHub) as the source of truth for Apex applications. The SQL representation of the application would be stored in a repository and manually deployed from there by DBAs.

After working for some time with Oracle Apex applications, developing shared code for AD group access, figuring out a way to do unit tests, etc. I determined that the no code / low code strategy was not profitable as applications become more and more complex. With the agreement of leadership, I spent some time researching alternatives, and arrived at the conclusion that the Django platform was a good solution for Simmons. This strategy was approved, and we started building our first full stack web applications with Django and deploying them to AWS.

2018-2020

Built Deployment Framework for Oracle BI Publisher Reports

Simmons Foods, Siloam Springs, AR

Leveraged Oracle BI Publisher APIs using a combination of Ruby and shell scripts whereby report definitions were stored in on-premise Git repositories and deployed to various BI Publisher environments (Development, QA, Production) using these tools. Eventually moved these repositories to GitHub and ran the deployments automatically using GitHub Actions and "on-premise" action runners living in AWS within a VPC that had access to our local network.

2018-2020

Built Deployment Framework for SQL Server Stored Procedures

Simmons Foods, Siloam Springs, AR

At Simmons, a third-party contractor had been hired to create reports within Oracle BI Publisher. The contractor had included raw SQL queries at the report level that were copied and pasted into the different reports with slight modifications to each query. I analyzed the code, developed a library of common stored procedures based on each use case, and rewrote each report to use the new stored procedures.

As with the BI Publisher reports, it was necessary to create a deployment framework that would use Git and later GitHub as the source of truth for the stored procedures, but would facilitate deployment to each tiered SQL server environment. Initially, the deployment process simply created a `RELEASE_FILE.sql` which we would hand off to the Infrastructure team. Later, we automated this step of the deployment using GitHub Actions and our "on-premise" action runners.

2018-2020

Simmons Foods, Siloam Springs, AR

Built Deployment Framework for Application Configuration Files

Oracle's PLM (Product Lifecycle Management) is a C#/.NET application which requires changing settings in a number of XML and KVP text files. When these files were under manual management, the software was very fragile and it was often impossible to track down the root cause of an instability.

I worked to establish Git / GitHub as a source of truth for these configuration files, and built a deployment framework using Ansible to not only push the files out to the servers in various environments, but also to manage scheduled jobs, install packages, and other OS-level tasks.